

RAHUL KUMAR

Software Development Engineer

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LinkedIn

Github

Pune, IN

EXPERIENCE

Software Development Engineer

Dice Enterprise Pvt. Ltd.

05/2024 – Present Pune, India

Smarter Procurement, Travel, Spend and Finance Management.

- Designed and developed a scalable multi-tenant payment orchestration system for enterprise reimbursements using Strategy and Factory design patterns to support client-specific workflows and provider routing.
- Built fault-tolerant payment workflows for IDFC FIRST Bank using CBS and CBDC integrations, including payment initiation, retry orchestration, fallback handling, reconciliation, and payment state management.
- Implemented distributed locking and idempotent transaction handling using Redis to prevent duplicate payment execution across multiple backend containers.
- Developed asynchronous retry and cron-based recovery pipelines for failed and pending payment transactions, improving reliability and operational recovery.
- Optimized backend processing by migrating AWS Lambda-based execution flows into centralized server-side orchestration systems, improving maintainability, observability, and scalability.
- Built high-volume report generation pipelines capable of streaming millions of MySQL records into compressed CSV exports with multipart AWS S3 uploads and automated email delivery workflows.
- Reduced report processing bottlenecks through SQL optimization, Ebean ORM tuning, manual pagination strategies, and parallel asynchronous processing using Vert.x.
- Integrated AWS S3, Google Cloud Storage, and Azure Blob Storage for secure file uploads, storage abstraction, and signed URL generation workflows.
- Developed enterprise-grade voucher settlement, GST reporting, payment reconciliation, and transaction tracking systems for large-scale financial workflows.
- Improved backend stability by debugging and resolving production issues involving payment failures, serialization issues, duplicate key conflicts, API latency, and distributed transaction inconsistencies.
- Reduced infrastructure and Google Maps costs by 50%, decreased monthly database costs by 20 %, and improved API performance and backend processing efficiency by 10%.

Software Engineer Intern

Nolan Edutech Pvt. Ltd

10/2023 – 03/2024 Bengaluru, India

- URL Shortener Platform Development:** Developed a scalable URL Shortener platform using Java, Spring Boot, MySQL, Redis, and Docker with support for custom aliases, link expiration, QR code generation, click analytics, and secure user authentication.
- REST API & Authentication Implementation:** Designed and implemented RESTful APIs for short URL creation, redirection handling, analytics tracking, and JWT-based authentication using Spring Boot.
- Base62 URL Generation System:** Designed a Base62-based URL generation flow using database-generated unique IDs to create compact, collision-free short links with efficient lookup and redirection mechanisms.
- Custom Alias & Redirect Optimization:** Developed customizable alias generation with uniqueness validation and optimized lookup mechanisms for personalized short URL creation while improving redirect response performance.
- Performance & Security Enhancements:** Integrated Redis caching and database indexing to reduce repetitive database lookups, optimize redirect latency, and efficiently handle 10K+ URL records. Implemented rate limiting and request validation to prevent API abuse.

SUMMARY

Backend-focused Software Engineer with experience building scalable payment orchestration systems, enterprise financial workflows, and high-volume backend processing pipelines using Java, Spring Boot, Vert.x, Redis, Kafka, MySQL, and cloud technologies. Skilled in distributed transaction handling, asynchronous processing, backend optimization, enterprise integrations, and fault-tolerant microservices for production-scale systems.

EDUCATION

Master of Computer Applications

2023 – 2025

Lovely Professional University, Punjab

B.Sc. in Mathematics (Honours)

09/2018 – 09/2021

C. M. Science College, Darbhanga, Bihar

Class 12th – Science

04/2016 – 04/2018

Kendriya Vidyalaya, Jawaharnagar, Sitamarhi, Bihar

SKILLS

Languages: Java, SQL, JavaScript

Backend Technologies: Spring Boot, Vert.x, Spring AI, REST APIs, JDBC, JPA, Hibernate, Microservices

Cloud & DevOps: AWS (EC2, Lambda, S3, CloudWatch), GCP, Docker, Kubernetes, Jenkins, CI/CD

Messaging & Caching: Kafka, Redis

Databases & Analytics: PostgreSQL, MySQL, MongoDB, RDS, Elasticsearch, BigQuery

Monitoring & Tools: Grafana, Kibana, Postman, GitHub, GitLab, JIRA

Concepts: API Security, Authentication & Authorization, Database Optimization, Transaction Management, System Design, Distributed Systems, Concurrency

CERTIFICATIONS

Full-Stack Web Development

04/2022 – 01/2023

Masai School, Bengaluru, Karnataka